

Lecture 4 Task: Real-Coded GA for Water Tank Design Optimization

Since we already solved the Rosenbrock function in class, this task uses a more practical engineering-style problem.

Task Title

Optimize the Dimensions of a Cylindrical Water Tank Using Real-Coded Genetic Algorithm

Problem Statement

A company wants to design a closed cylindrical water tank that can store at least:

$$V_{\min} = 1 \text{ m}^3$$

The tank has two design variables:

r = tank radius

h = tank height

The goal is to find the best values of r and h such that the total material cost is minimized while satisfying the required volume constraint.

Design Variables and Bounds

Variable	Meaning	Lower Bound	Upper Bound
r	Tank radius	0.1 m	1.5 m
h	Tank height	0.2 m	4.0 m

So each chromosome is a real-valued vector:

$$X = [r, h]$$

Volume Constraint

The volume of the cylinder is:

$$V = \pi r^2 h$$

The tank must satisfy:

$$V \geq 1 \text{ m}^3$$

Cost Function

Assume the tank is closed from the top and bottom.

The surface area of the cylindrical side is:

$$A_{\text{side}} = 2\pi rh$$

The area of the top and bottom is:

$$A_{\text{top+bottom}} = 2\pi r^2$$

Assume the following material costs:

Part	Material Cost
Side wall	Rs. 500 per m ²
Top and bottom	Rs. 700 per m ²

So the material cost is:

$$C = 500(2\pi rh) + 700(2\pi r^2)$$

Because the tank must have at least 1 m³ volume, use a penalty if the volume is less than 1:

$$\text{Penalty} = \begin{cases} 100000(1 - V)^2, & V < 1 \\ 0, & V \geq 1 \end{cases}$$

Therefore, the final objective function is:

$$F = C + \text{Penalty}$$

The goal is:

$$\min F$$

Required GA Settings

Use a Real-Coded Genetic Algorithm with the following settings:

Parameter	Value
Population size	20
Number of generations	50
Number of variables	2
Selection	Binary tournament selection
Crossover	Simulated Binary Crossover, SBX
Crossover probability p_c	0.9
SBX index η_c	15
Mutation	Polynomial mutation
Mutation probability per variable p_m	$\frac{1}{2}$
Mutation index η_m	20
Survivor selection	Elitist strategy

Students may use DEAP or write their own code.

Hand Calculation Requirement

Before writing the complete program, perform **one full iteration by hand** using:

Population size = 4

Randomly choose the initial values of r and h within the given bounds. For these 4 individuals, show:

1. Initial population values
2. Volume of each tank
3. Cost of each tank
4. Penalty value
5. Final objective value
6. Binary tournament selection result
7. Mating pool
8. One crossover and mutation step
9. Updated population after one iteration

Required Output

Your program should print the final best solution:

Quantity	Value
Best radius r	
Best height h	
Tank volume V	
Material cost C	
Penalty	
Final objective value F	

Also show whether the final solution satisfies:

$$V \geq 1 \text{ m}^3$$

Required Tables

Table 1: Initial Population

Show the initial population:

Index	r	h	Volume	Cost	Penalty	Objective
1						
2						
3						
...						
20						

Table 2: Final Population

Show the final population after 50 generations:

Index	r	h	Volume	Cost	Penalty	Objective
1						
2						
3						
...						
20						

Table 3: Best Solution History

Record the best objective value after each generation:

Generation	Best F
1	
2	
...	
50	

Questions to Answer

1. Why is real-coded GA suitable for this problem?
2. What happens if the tank volume is less than 1 m³?
3. Why did we add a penalty term?
4. What is the role of SBX crossover in this problem?
5. What is the role of polynomial mutation?
6. Did the final solution satisfy the volume constraint?
7. Is the best tank short and wide, or tall and narrow? Explain why.

Bonus Task

Run the algorithm three times with different random seeds.

Compare the results:

Run	Best r	Best h	Volume	Cost	Objective
1					
2					
3					

Then answer:

Does the GA always find exactly the same solution? Why or why not?